

Digital Innovation

What is Digital Innovation?

- Use of digital technologies to **create new or improved** products, processes, services, and business models.
- **Products** → What users interact with (apps, features)
- **Processes** → How work is done internally (delivery, payments, operations)
- **Services** → How customers are served (speed, convenience, personalization)
- **Business models** → How money is made (subscriptions, commissions, platforms)

“Digital innovation can change *what* you offer, *how* you operate, or *how* you earn.”

What is Digital Innovation?

Driven by solving real problems and creating value

- Problem: Long queues at banks
- Digital innovation: Mobile banking apps

Example:

- “In Nepal, people often had to travel far to pay bills. Apps like eSewa solved a real pain point — time, travel, and inconvenience.”

Cause–Effect:

- Real problem → Digital solution → User adoption → Business growth

“Technology alone does not create value — solving real problems does.”

What is Digital Innovation?

Continuous and cumulative — small improvements matter too

- Innovation doesn't always mean a big breakthrough
- Small changes add up over time

Example:

- Pathao adding:
 - In-app wallet
 - Location sharing
 - Ride ratings

“Each small update improves experience. Over time, these small improvements create major competitive advantage.”

What is Digital Innovation?

Not just **tech adoption** — it's meaningful change in how organizations and users operate and interact.

- Simply having a website ≠ innovation
- Using data to personalize services = innovation

Context:

- “Many businesses in Nepal created Facebook pages during COVID. Only some used digital tools to truly change how they served customers.”

“Innovation is about changing behavior, not just installing software.”

What is Digital Innovation?

- **Innovation ≠ invention**
Innovation = applying technology to create value

Examples:

- Smartphones were not invented by WhatsApp
- Ride-sharing was not invented by Pathao

But:

- “Pathao innovated by applying GPS, mobile internet, and digital payments in a Nepali context.”

Why Digital Innovation Matters

- Improves productivity and efficiency
- Enhances customer experience
- Builds competitive advantage
- Enables new revenue streams
- Supports data-driven decision-making

Types of Digital Innovation

- **Product Innovation** — new or improved digital products
- **Process Innovation** — internal workflows become efficient
- **Business Model Innovation** — new ways of value creation
- **Service / Customer Experience Innovation** — better user interactions
- **Marketing Innovation** — digital channels & targeted reach
- **Organizational Innovation** — culture and structure change

Examples of Digital Innovation

Netflix streaming

- Types of Innovation:
 - Product Innovation → Streaming platform instead of DVDs
 - Business Model Innovation → Subscription-based model

Amazon AWS

- Types of Innovation:
 - Product Innovation → Cloud computing services
 - Process Innovation → Efficient, scalable infrastructure for businesses

AI chatbots and automation

- Types of Innovation:
 - Service / Customer Experience Innovation → 24/7 customer support and task automation

Examples of Digital Innovation

- **Daraz logistics optimization**
- Types of Innovation:
 - Process Innovation → Faster, efficient order delivery
- **Pathao ride-sharing efficiency**
- Types of Innovation:
 - Process Innovation → Efficient driver-rider matching
 - Service / Customer Experience Innovation → Better ride experience
- **eSewa adding bill payments & ticketing**
- Types of Innovation:
 - Product Innovation → New features in app
 - Service / Customer Experience Innovation → More convenient services for users

Process Innovation

- “Improving how things are done using digital tools.”

- **Examples:**

- Automation
- Optimization
- Faster decision-making

- **Nepali Example – Daraz Logistics**

- Orders tracked digitally
- Delivery routes optimized
- Warehouse operations automated

“The customer sees faster delivery, but the real innovation is behind the scenes.”

Product Innovation

- “Creating new or significantly improved digital products.”
- New features
- Better user experience
- Personalization
- **Nepali Example – eSewa**
- Started as simple payment app
- Added mobile recharge, bill payments, ticketing
- **Global Example**
- Spotify playlists
- Netflix recommendations

Business Model Innovation

“Changing how a company makes money.”

- Not just selling products
- Using platforms, subscriptions, commissions
- **Nepali Example – Pathao**
- Does not own vehicles
- Earns commission per ride
- Uses data to optimize matching

“Pathao is not a transport company — it is a digital platform.”

- **Another Example**
- Foodmandu taking commission from restaurants
- Daraz earning from sellers, ads, logistics

Service / Customer Experience Innovation

- Focus on **making life easier for users**

Nepali Examples:

- **Pathao:** GPS tracking, driver ratings, real-time updates
- **eSewa:** Instant notifications, easy bill payments, seamless top-ups

Global Examples:

- **Netflix:** Personalized content recommendations
- **Amazon:** One-click purchase and delivery tracking

“The way a user experiences a service often determines adoption more than the technology itself.”

Marketing Innovation

- Focus on **reaching the right people with digital tools**
- **Nepali Examples:**
- **Daraz:** Targeted flash sale campaigns via SMS, app notifications, and social media
- **Foodmandu:** Promoting restaurant partners with app-based notifications
- **Global Examples:**
- **Spotify:** Targeted playlists and ads
- **Coca-Cola:** Personalized campaigns using social media analytics

Organizational Innovation

- Focus on **how the company works internally** to support digital growth

Small Teams Launching Features Quickly

- **Daraz Nepal**
- Small product teams work on app updates like **flash sale features**.
- Can test and release a new feature in **days instead of months..**

Cross-Team Collaboration for Faster Problem Solving

- **Foodmandu**
- Logistics team + restaurant relations + customer support coordinate to **resolve delivery delays** quickly.

Theoretical Example:

- Startup faces payment failures.
- Finance team + app developers + QA team collaborate for a **same-day fix**, instead of waiting weeks.

Technologies Driving Digital Innovation

Artificial Intelligence (AI)

“AI allows systems to learn from data and make decisions.”

Uses:

- Recommendations
- Fraud detection
- Demand prediction

Nepali Context:

- Ride demand prediction
- Banking fraud detection

Technologies Driving Digital Innovation

Internet of Things (IoT)

“IoT connects physical objects to the internet.”

Examples:

- GPS in delivery bikes

Nepali Use Case:

- Pathao tracking riders
- Logistics tracking

Technologies Driving Digital Innovation

Cloud Computing

- “Cloud allows businesses to rent computing power instead of owning it.”

Nepali Impact:

- Low startup cost
- Easy scaling

Example:

- Daraz flash sales handled via cloud scaling

Technologies Driving Digital Innovation

Big Data

- “Big data helps companies understand user behavior.”

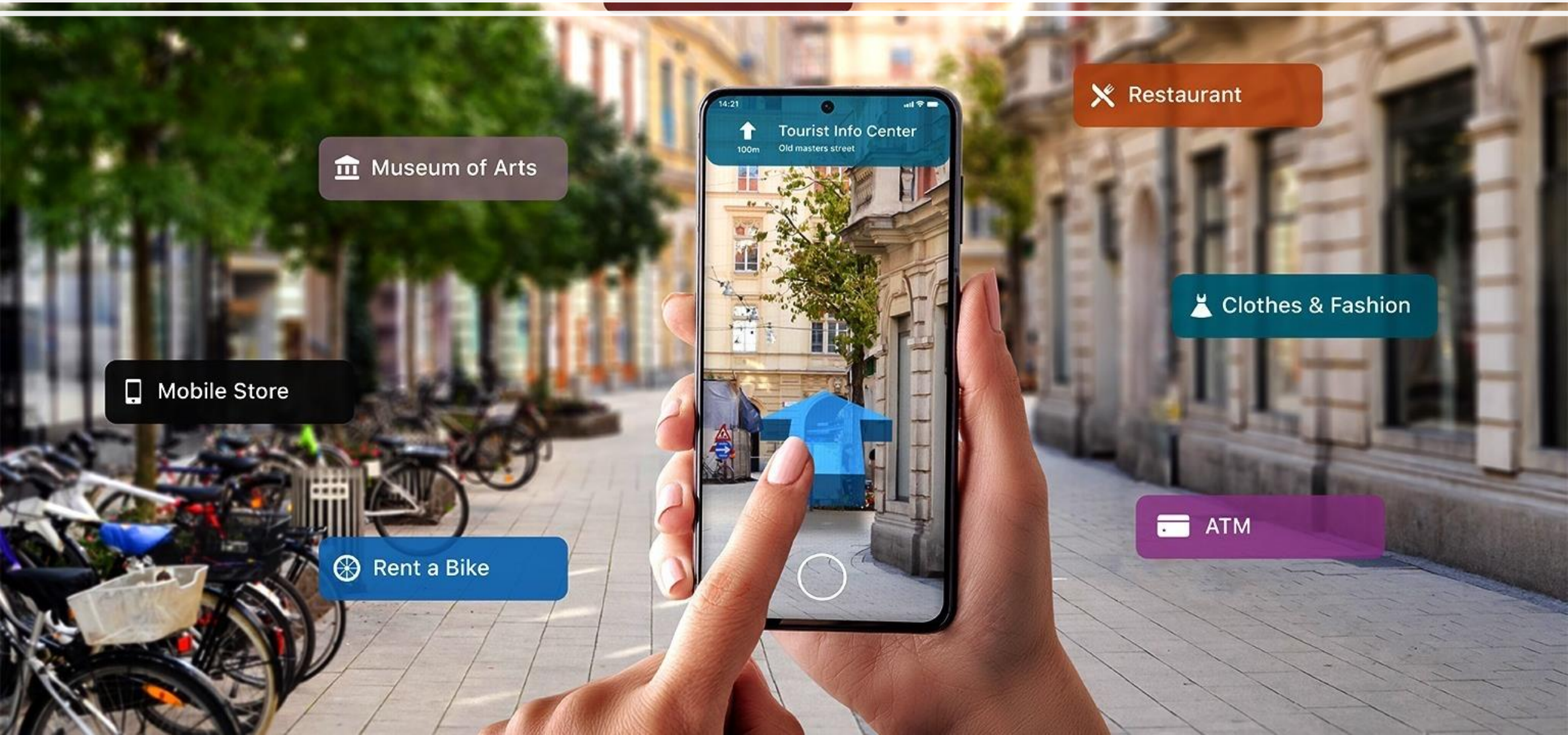
Nepali Examples:

- Daraz analyzing purchase patterns
- Pathao optimizing routes

Technologies Driving Digital Innovation

- **AR (Augmented Reality)**
 - Adds digital elements to the real world
 - **Innovation use:** Enhances customer experience & marketing
 - **Examples:**
 - IKEA Place app → Visualize furniture at home
 - Snapchat filters → Interactive marketing campaigns
- **VR (Virtual Reality)**
 - Creates fully digital immersive environments
 - **Innovation use:** Improves training, education, and service experiences
 - **Examples:**
 - Virtual real estate tours → Explore homes remotely
 - VR medical training → Practice surgeries safely

Augmented Reality



 Museum of Arts

 Mobile Store

 Rent a Bike

 Restaurant

 Clothes & Fashion

 ATM



Virtual Reality

Innovation Process & Culture

- **Stages:** Ideation → Prototype/MVP → Scale
- **Ideation**
 - Generate ideas to solve a problem or create value
 - Example: Brainstorming ways to make online food delivery faster
- **Prototype / MVP (Minimum Viable Product)**
 - Build a simple version of the product to test the idea
 - Example: Launching a basic app to accept orders before adding advanced features
- **Scale**
 - Improve and expand the product after testing and feedback
 - Example: Add multiple restaurants, GPS tracking, digital payments, and wider delivery areas

Innovation Process & Culture

- **Key Principles**
- **Experimentation and feedback**
 - Test ideas quickly, learn from real users, and improve
 - Example: Collect customer ratings on app usability and delivery speed
- **Collaboration across teams**
 - Work together—tech, marketing, operations, and design—to make innovation successful
 - Example: Developers build the app, marketing promotes it, operations handle logistics

Challenges of Digital Innovation

- **Resistance to Change**
 - People may prefer old ways of working
 - Example: Employees reluctant to adopt a new app-based ordering system
- **Skills Gaps**
 - Lack of technical or digital skills to implement innovation
 - Example: Small businesses struggling with digital marketing or app management
- **Data Security & Privacy**
 - Protecting customer and business data is critical
 - Example: Online payments or customer databases can be vulnerable to breaches
- **High Initial Investment**
 - Innovation may require money for technology, infrastructure, or training
 - Example: Building a mobile app or buying VR equipment
- **Measuring ROI (Return on Investment)**
 - Hard to know if innovation brings real value or profit
 - Example: Spending on AR marketing may not immediately increase sales

Challenges of Digital Innovation

“Even good ideas can fail **without the right culture, skills, and governance.**”

- **Culture:** Employees must be open to change
- **Skills:** Teams must know how to use technology effectively
- **Governance:** Clear policies and oversight ensure security and ROI

Challenges of Digital Innovation in Nepal

- **Limited Infrastructure**

- Internet speed, electricity, or hardware may be unreliable
- Example: Online education struggles in rural areas due to poor connectivity

- **Digital Skills Gap**

- Lack of technical know-how among users or employees
- Example: Small shops unable to manage e-commerce platforms effectively

- **Trust and Adoption Issues**

- People may hesitate to use digital services due to fear, habits, or skepticism
- Example: Reluctance to use digital payments instead of cash

“Innovation is not only about technology — it is about context.”

Nepal's Innovation Context

- **Limited Infrastructure**

- Internet and electricity constraints affect digital adoption
- Example: Online learning apps must work on low bandwidth

Emerging Opportunities

- **Fintech:** Digital payments, microloans, mobile wallets
 - Example: **eSewa, Khalti** enabling cashless transactions
- **Agri-tech:** Farm advisory, crop tracking, and supply chain solutions
 - Example: Apps connecting farmers to markets

Mobile Penetration is Essential

- Smartphones are the main access point for digital services
- Example: Ride-sharing apps like **Pathao** rely on mobile users

Startups Innovate by Adapting Global Ideas Locally

- Not inventing from scratch, but tailoring global tech to Nepal's context

Key Takeaways

- Innovation is multi-faceted
- Fueled by tech and culture
- Essential for competitiveness
- Nepal is innovating in unique ways